

LEARNYMO

SOFTWARE REQUIREMENTS

SPECIFICATION

Project Title: LearnMo – Online Learning Marketplace

Tagline: “Your Topic. Your Time. Your Teacher.”

Derived from the LearnMo Complete MVP Development Specification

1. Project Overview

1.1 Introduction

LearnMo is an online learning platform connecting students with teachers based on the student's exact learning requirement. Students can learn a specific topic, a complete unit, or a full subject/syllabus. The MVP supports 1-to-1 classes, group classes, paper classes, unit classes, topic classes, full-subject classes and class series.

1.2 Problem Statement

LearnMo addresses the difficulty of finding a suitable teacher for an exact learning requirement, choosing a suitable time, requesting or discovering classes, and managing the learning process in one platform.

1.3 Proposed Solution

The platform connects Student Need → Request/Demand → Teacher → Class → Booking → Payment → Chat → Online Class → Feedback.

2. Project Objectives

- Enable topic-, unit- and subject-level learning requests and class discovery.
- Allow preferred dates, time ranges and durations.
- Allow teachers to define teaching areas, fees and availability.
- Provide teacher matching and a demand system.
- Support booking/enrollment, payments, chat and online classroom access.
- Provide syllabus progress and study-planning features according to MVP priority.
- Provide administration tools for users, teachers, academic data, classes, demand, payments and content.

3. Target Users

User Type	Description
Student	Manages academic information, searches/requests classes, books or enrolls, pays, chats, attends and tracks progress.
Teacher	Registers, submits verification documents, creates teaching profile, defines subjects/topics, fees and availability, responds to demand and conducts classes.
Admin	Manages students, teachers, verification, academic structure, classes, demand, payments, content and

	platform settings.
Guest User	Accesses public website content without logging in.

4. Project Scope

- Public website and information pages.
- Student registration, login, academic profile and dashboard.
- Teacher registration, verification, profile and dashboard.
- Admin dashboard.
- Dynamic academic hierarchy: Grade → Stream → Subject → Unit → Sub-unit → Topic.
- 1-to-1 requests, group, paper, unit, topic, full-subject and series classes.
- Demand recording and teacher matching.
- Booking/enrollment and booking statuses.
- Payment, wallet and teacher earnings.
- Platform chat and class cards.
- Online classroom with controlled access.
- Timetable, notifications, syllabus/progress, study planner and learning resources as specified.

5. Website Modules

Module	Description
Public Website	Home, About, How It Works, For Students, For Teachers, Contact, FAQ, Terms and Privacy.
User Management	Registration, login, profile and role management.
Class Booking	1-to-1 requests, group enrollment and class categories.
Demand & Matching	Records student needs and exposes relevant demand to teachers.
Teacher Management	Registration, verification, teaching profile, fees, availability and schedule.
Class Management	Creation, publishing, enrollment, status and series.
Chat	Booking-linked text chat with class card and Join Class action.
Online Classroom	Authorized student/teacher video and audio class access.
Payments & Wallet	Class/membership payments, earnings, withdrawals and transactions.
Syllabus & Study	Topic status, recommendations and 1–4–7 study planner.
Content	Past papers, notes and learning resources.
Notifications	Booking, payment, chat, class, demand, cancellation and feedback events.
Admin	Users, teachers, academic structure, classes, demand, payments and content.

6. Website Pages

- Home
- About LearnMo
- How It Works

- For Students
- For Teachers
- Contact Us
- FAQ
- Terms & Conditions
- Privacy Policy
- Student Dashboard
- Class Booking
- My Requests
- My Classes
- Timetable
- Chat
- My Syllabus & Progress
- Study Planner
- Past Papers & Notes
- Wallet
- Profile
- Notifications
- Settings
- Teacher Dashboard
- My Classes
- Create Class
- Demand
- Schedule
- Wallet
- My Profile
- Chat
- Notifications
- Settings
- Help & Support
- Admin Dashboard
- Students
- Teachers
- Teacher Verification
- Classes
- Demand
- Academic Management
- Payments
- Past Papers & Notes
- Reports
- Notifications
- Settings

7. Functional Requirements

ID	Requirement	Description
FR-01	Student Registration	Register with required

		personal/academic information and generate a unique Student ID.
FR-02	Teacher Registration	Register and submit verification information/documents.
FR-03	Teacher Verification	Admin reviews, approves or rejects teacher verification.
FR-04	Academic Structure	Admin manages Grade → Stream → Subject → Unit → Sub-unit → Topic dynamically.
FR-05	1-to-1 Request	Student selects academic information, requirement, medium, date, time range, duration and payment preference, then submits.
FR-06	Teacher Matching	System uses teaching configuration and availability to support matching.
FR-07	Group Classes	Teachers create group classes; students browse, view and enroll.
FR-08	Demand	Topic, unit, paper and group-class needs are recorded as demand for relevant teachers.
FR-09	Class Creation	Verified teachers create supported class types with academic, schedule, fee and capacity data.
FR-10	Booking Status	Support Pending, Teacher Found, Teacher Accepted, Student Accepted, Payment Pending, Confirmed, Completed, Rejected and Cancelled.
FR-11	Payment	Support class/group/membership payments and teacher earnings/withdrawals.
FR-12	Chat	Enable booking-linked chat after confirmation.
FR-13	Online Classroom	Only users associated with a confirmed booking can access the classroom.
FR-14	Timetable	Only confirmed classes appear in confirmed timetables.
FR-15	Progress	Students mark topics Not Started, Not Clear or Clear.
FR-16	Recommendations	Matching classes for Not Clear topics are recommended when available.
FR-17	Study Planner	Support the specified 1–4–7 study rule and preferred study slots.
FR-18	Ratings	Students rate completed classes 1–5 stars with optional written feedback.
FR-19	Notifications	Provide relevant booking, payment, chat, class, demand and feedback notifications.
FR-20	Admin Management	Admin manages users, teachers, academic data, classes, demand, payments and content.

8. Major System Workflows

8.1 Primary 1-to-1 Workflow

Student → Grade → Subject → Topic/Unit/Full Subject → Medium → Preferred Date → Preferred Time Range → Duration → Submit Request → Teacher Matching → Teacher Fee & Availability → Student Accepts → Payment → Booking Confirmed → Chat → Online Class → Completion → Rating.

8.2 Group Demand Workflow

Student Need → Demand Recorded → Teacher Sees Demand → Teacher Creates Class → Matching Students Notified → Students Enroll.

8.3 Personalized Learning Workflow

Student Studies Topic → Marks Not Clear → LearnyMo Records Need → Teacher Creates Matching Class → System Detects Match → Student Notification → Book/Enroll → Learn → Mark Topic Clear.

9. User Roles and Permissions

Role	Key Permissions
Student	Own profile/academic data; requests; browse/book/enroll; payments; own chat/classes; progress.
Teacher	Own profile; qualifications; teaching areas; fees; availability; classes; relevant demand; classes; earnings.
Admin	Users; teachers; verification; academic hierarchy; classes; demand; payments; content; settings.

10. Non-Functional Requirements

- Security: secure authentication, password hashing, RBAC, authorization, secure document storage, payment security, upload validation and basic audit logs.
- Access control: users must not access another user's booking, chat or classroom.
- Responsive design: desktop, laptop, tablet and mobile.
- Usability: modern, clean, professional, educational, Gen-Z friendly, easy to use and not childish.
- Maintainability: academic data must be dynamic rather than hard-coded.
- Classroom access must be protected by confirmed booking state.

11. UI/UX Requirements

- Use official LearnyMo branding: Blue + Teal/Green + Purple + White + Yellow accents.
- Use the official LearnyMo logo without modifying its design.
- Prioritize mobile screens: booking, class details, chat, join class, teacher class creation, demand, timetable and wallet.
- Use clear class cards showing teacher, academic information, date/time, duration, fee and capacity.
- Provide a dedicated Join Class action in relevant chat class cards.

12. Technology Requirements

Area	Technology
Frontend	React + TypeScript + Vite
Styling	Tailwind CSS
Backend	Node.js + Express.js + TypeScript
Database	MongoDB Atlas
ODM	Mongoose
Authentication	JWT + Refresh Tokens
API	REST API
Realtime	Socket.IO
File Storage	Cloudflare R2
Images	Cloudinary
Payments	PayHere
Classroom	LiveKit
AI	Gemini API
Email	Resend
Search	MongoDB Atlas Search
Push Notifications	Firebase Cloud Messaging
Caching/Queues	Redis + BullMQ when needed
Monitoring	Sentry
Analytics	Google Analytics
Frontend Hosting	Vercel
Backend Hosting	Render
Source Control	GitHub
Deployment	Docker

13. Database Requirements

- User → Student / Teacher profiles.
- Academic: Grade → Stream → Subject → Unit → Sub-unit → Topic.
- Learning: Student → Request/Demand → Teacher Matching → Class → Booking → Payment → Chat → Online Classroom → Completion → Rating.
- Wallet and transaction records for students and teachers.
- Booking, notification and class status records.
- Past papers, notes and learning-resource metadata.

14. API Requirements

- Authentication and profile APIs.
- Student academic profile and progress APIs.
- Teacher profile, verification, teaching configuration and availability APIs.
- Academic hierarchy APIs.
- Class creation, discovery and enrollment APIs.
- 1-to-1 request and matching APIs.
- Demand APIs.
- Booking and status-transition APIs.
- Payment, wallet, earnings and withdrawal APIs.
- Chat and notification APIs.
- Online classroom authorization/access APIs.

- Content APIs.
- Admin management/reporting APIs.

15. Data Requirements

- Student: personal/academic information, requests, bookings, progress, wallet transactions and feedback.
- Teacher: personal information, qualifications, verification, teaching areas, fees, availability, classes, demand, earnings and ratings.
- Class: type, academic hierarchy, teacher, schedule, duration, fee, capacity, enrollment and status.
- Booking: student, teacher, class/request, status, payment state and timestamps.
- Demand: academic requirement, interested students and preferred time.
- Content: academic metadata and file/resource information.

16. Constraints and Assumptions

- First production target is a responsive web platform.
- Native Android/iOS are future scope, not first MVP requirements.
- Advanced chat, advanced classroom, whiteboard, attendance, recordings, AI tutor/quiz/recommendations and advanced analytics should not delay first MVP.
- The source specifies a suitable Sri Lankan payment gateway; implementation details are selected during development.
- Final legal terms and non-circumvention clauses require legal review.
- The source specifies an 80% Teacher / 20% LearnMo revenue split.

17. Testing Requirements

- Unit testing of critical services and business rules.
- API testing for authentication, authorization, booking, payment and class flows.
- Role/permission testing.
- Booking state-transition testing.
- Payment and wallet testing.
- Chat and classroom authorization testing.
- Responsive UI testing.
- File upload/validation testing.
- End-to-end testing from class request through rating.
- Security testing against unauthorized access to another user's booking, chat or classroom.

18. MVP and Future Scope

18.1 P0 – Must Have

- Student registration/login, academic profile, class booking, 1-to-1 requests, group classes/enrollment, My Requests, timetable, payment, chat, class link and online classroom.
- Teacher registration/login, verification, profile, grades, subjects, units, topics, fees, availability, classes, poster, demand, schedule, wallet, chat and online classroom.
- Admin user management, teacher verification, academic management, class management, demand management and payment management.

18.2 P1 – Important

- My Syllabus and Not Started / Not Clear / Clear.
- Not Clear → Class Recommendation.
- Study Planner and 1–4–7 rule.
- Past Papers & Notes.
- Teacher Rating and Repeat Rate.
- Notifications, Class Series, Paper, Unit, Topic and Full Subject Classes.

18.3 Phase 2 / Future

- Advanced chat
- Advanced video classroom
- Online whiteboard
- Attendance
- Recorded classes
- AI Quiz
- AI Tutor
- AI recommendations
- Advanced analytics
- Native Android app
- Native iOS app

19. Project Deliverables

- Responsive public website.
- Student application.
- Teacher application.
- Admin dashboard.
- REST API backend.
- MongoDB database and academic hierarchy.
- Authentication and role-based authorization.
- Booking, demand and matching modules.
- Payment and wallet modules.
- Chat and online classroom integration.
- Notification system.
- Testing and deployment configuration.
- Technical documentation.

20. Project Timeline

The source requirements template contains a timeline section, but the supplied LearnMo specification does not define dated milestones. Therefore, no specific duration is assumed. The implementation order below follows the specified MVP priorities.

Phase	Focus
1	Project setup, authentication, roles, academic hierarchy and database/API foundation.
2	Student/teacher profiles, class creation/discovery and 1-to-1 requests.

3	Demand, matching, booking states and timetable.
4	Payment, wallet, earnings and transactions.
5	Chat, class cards and online classroom.
6	P1 features: syllabus, recommendations, planner, resources, ratings and notifications.
7	Testing, security, responsive QA, deployment and launch preparation.

Final Functional Backbone

Student requests a class → Teacher is matched → Teacher accepts → Student accepts → Payment → Booking confirmed → Chat opens → Class card appears → Student/Teacher click Join Class → Online class → Completion → Rating.

Source basis: this document follows the uploaded Website Software Requirements Document structure and derives its LearnMo requirements from the uploaded Complete MVP Development Specification. Where the source did not provide a dated timeline or implementation detail, it has not been presented as a confirmed requirement.